

Supervised Learning Assignment — Proposal (Submission 1)

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Project title

Supervised regression for global movie revenue prediction (1950–2026) using a tabular Kaggle dataset.

Dataset

Name (Kaggle): Global Movies Dataset (1950–2026)
Source link (public): <https://www.kaggle.com/datasets/suhanigupta04/global-movies-dataset-19502026>
File used in code: `global_movies_dataset_1950_2026.csv`

Target variable

The supervised learning target is `revenue_million`: total revenue in millions (continuous numeric outcome).

Task type

Regression (evaluate with MAE, RMSE, and R^2).

Problem statement

This project predicts movie revenue from production/marketing signals and audience/critic indicators. We compare multiple supervised models, test preprocessing choices, evaluate robustness, and interpret feature importance to support a practical recommendation.

Research questions (7)

1. Baseline performance: How well do Linear Regression, Decision Tree, and k-NN predict revenue on a held-out test set?
2. Model comparison: Which models (Ridge, Random Forest, Gradient Boosting, Linear SVR, optional XGBoost) achieve the best MAE / RMSE / R^2 ?
3. Preprocessing effect: How do preprocessing strategies affect a fixed Random Forest pipeline?
4. Feature importance: Which variables contribute most to a tree-based model's predictions?
5. Metric sensitivity: Do model rankings change when using MAE vs RMSE vs R^2 ?
6. Robustness: How stable is performance under cross-validation and mild numeric noise?
7. Practical recommendation: What is the best trade-off among accuracy, interpretability, robustness, and deployment cost?

Models (planned/used)

- LinearRegression, Ridge
- DecisionTreeRegressor, KNeighborsRegressor
- RandomForestRegressor, GradientBoostingRegressor
- LinearSVR (with scaling)
- XGBRegressor (optional if installed)

Evaluation metrics

- MAE • RMSE • R^2
- Robustness: cross-validated RMSE (mean $\pm$ std)

Expected figures and tables

- Table I / Figure 1: Baseline model MAE, RMSE, R^2 and grouped bar chart.
- Table II / Figure 2: Full model comparison; bar chart of test R^2 .
- Table III / Figure 3: Preprocessing ablation (Random Forest).
- Table IV / Figure 4: Top feature importances (bar plot).
- Table V / Figure 5: Model ranks under different metrics.
- Table VI / Figure 6: CV RMSE bar chart; noise-perturbation summary.
- Table VII / Figure 7: Qualitative decision matrix and radar figure.

Current outputs snapshot (see ML/outputs folder)

RQ1 best baseline by RMSE: LinearRegression (RMSE=164.168, MAE=92.017, R^2 =0.343)

RQ2 best overall by RMSE: GradientBoosting (RMSE=157.262, MAE=83.468, R^2 =0.397)

RQ6 best CV RMSE: RandomForest (mean=152.649, std=4.345)